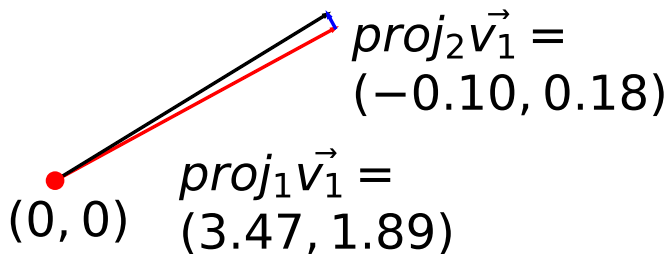


first and second principal components of $\vec{v}_1$

$$\vec{v}_1 = (3.38, 2.07)$$



$$\|\vec{v}_1\|^2 = \|proj_1 \vec{v}_1\|^2 + \|proj_2 \vec{v}_1\|^2 = 3.47^2 + 1.89^2 + (-0.10)^2 + 0.18^2$$